

Banking Transaction Data Pipeline - Week 1

Data Quality & Profiling Report

1. Dataset Overview

- Total Number of Records (Rows): 300000
- Total Number of Attributes (Columns): 10
- Target Output File: master_transactions.csv

2. Data Quality Assessment Summary

Data Quality Check Item	Metric Value	Status / Remarks
Total Duplicate Rows	0	Clean
Duplicate Transaction IDs	0	Clean
Missing Customer IDs	592	Orphaned transaction risk
Missing Account Numbers	607	Incomplete ledger record
Negative Amount Transactions	865	Business rule violation
High Value Transactions (>10k)	149249	Flagged for risk inspection
transaction_datetime Type	str	Mismatched Schema (Needs Cast)
account_number Type	float64	Incorrect Float Format

3. Key Observation:

- Wrong Date Format:** The transaction_datetime is saved as regular text (str), making it impossible to filter or sort transactions by date.
- Bad Account Numbers:** The account_number is saved as a decimal (float64), which accidentally adds a .0 at the end and deletes important zeros from the front.
- Blank Identity Fields:** There are **592 transactions missing Customer IDs** and **607 transactions missing Account Numbers**, which makes these records incomplete.
- Illegal Negative Amounts:** There are **865 transactions with negative values**, which breaks standard banking rules for normal transactions.

- e. **High-Risk Fraud Flags:** A massive chunk of **149,249 transactions is over \$10,000**, which must be flagged and monitored closely for security and fraud inspection.

4. Recommendations for Data Cleaning (ETL Stage)

- a. **Account Number Fix:** Account numbers are identifiers, not math values, so they must always be converted into **Strings (Object type)** to preserve their exact formatting.
- b. **Timestamp Casting:** Convert transaction_datetime text strings into native **Python Datetime objects** using pandas.to_datetime() to enable time-based tracking.
- c. **Deduplication Strategy:** Drop complete duplicate records and flag repeating transaction_id keys to ensure data uniqueness in the database.
- d. **Business Rule Enforcement:** Convert negative transaction numbers to absolute positive values or isolate them in a separate quarantine table for audit.
- e. **Missing Value Handling:** Fill blank Customer IDs and Account Numbers with default placeholder values like UNKNOWN_CUST to prevent database relational errors.